

HR Attrition & Workforce Analytics

SQL & Power BI

A complete walkthrough of database modeling, exploratory data analysis, SQL scripting, and interactive Power BI dashboard reporting for your portfolio presentation.

Project Context & Objectives



The Data Structure

Designed a relational data schema consisting of **fact-attrition** and **dim-employee** tables to evaluate key operational metrics, departmental retention trends, and employee demographics.



Core Business Questions

1. What are the key factors driving employee attrition? (Covers Q1, Q2, Q3, Q4, Q5, and Q6).
2. Which departments, Job roles, and employee segment have the highest attrition risk? (Covers Q7, Q8 and Q9).
3. How do employee satisfaction, work-life balance, tenure, income, and age relate to attrition? (Covers Q10, and Q11.)



Technical Stack

Utilized **MySQL** for complex relational aggregations and **Power BI** for executive-ready interactive visual storytelling.

Query 1: Overall attrition rate

- Measures overall employee attrition by calculating total employees who left and the overall attrition rate.
- Provides a baseline metric to understand the organization's overall employee turnover

```
select count(*) as Total_Attrition,  
sum(Attrition = 'Yes') as Attrition,  
round(100 * Avg(Attrition = 'Yes') , 1 ) as Attrition_rate_pct  
from fact_attrition;
```

	Total_Attrition	Attrition	Attrition_rate_pct
▶	1000	203	20.3

Query 2: Attrition rate by Overtime

- Compares employee attrition rates between employees who work overtime and those who do not.
- Identifies whether overtime is associated with higher employee turnover.

```
select OverTime,  
count(*) as Headcount,  
round(100 * AVG(Attrition = 'Yes'),1) as Attrition_rate_pct  
from fact_attrition  
group by OverTime  
order by Attrition_rate_pct desc;
```

	OverTime	Headcount	Attrition_rate_pct
▶	Yes	280	31.8
	No	720	15.8

Query 3: Attrition Rate by Age Group

- Compares employee attrition rates across age groups
- Helps identify which age groups have higher employee turnover

```
select AgeGroup ,  
count(*) as Headcount,  
round(100 * avg(Attrition = 'Yes'),1) as Attrition_rate_pct  
from fact_attrition f  
join dim_employee e on e.EmployeeID = f.EmployeeID  
group by AgeGroup  
order by field(e.AgeGroup, 18-25, 26-35, 35-45, 45-55, 56-60);
```

	AgeGroup	Headcount	Attrition_rate_pct
▶	36-45	398	19.6
	26-35	347	22.2
	46-55	141	9.9
	18-25	95	35.8
	56-60	19	0.0

Query 4: Attrition Rate by Salary Band

- Compares employee attrition rates across different salary bands.
- Identifies which salary band has the highest employee turnover.

```
select SalaryBand,  
Count(*) as Headcount,  
round(100 * avg(Attrition = 'Yes'), 1) as Attrition_rate_pct  
from fact_attrition  
group by SalaryBand  
order by field(SalaryBand, 'Low', 'Medium', 'High', 'Very High');
```

	SalaryBand	Headcount	Attrition_rate_pct
►	Low	250	25.2
	Medium	251	25.5
	High	249	22.1
	Very High	250	8.4

Query 5: Attrition Rate by Tenure Band

- Compares employee attrition rates across different tenure bands.
- Identifies which tenure group has the highest employee turnover.

```
select TenureBand,  
Count(*) as Headcount,  
round(100 * avg(Attrition = 'Yes'),1) as Attrition_rate_pct  
from fact_attrition  
group by TenureBand  
order by field(TenureBand, '0-2 yrs', '3-5 yrs', '6-10 yrs', '10+ yrs');
```

	TenureBand	Headcount	Attrition_rate_pct
►	0-2 yrs	525	26.5
	3-5 yrs	232	18.5
	6-10 yrs	158	10.1
	10+ yrs	85	5.9

Query 6: Overtime x Salary Band

- Analyzes attrition by combining overtime status and salary band.
- Identifies high-risk combinations where employees are more likely to leave.

```
Select  
SalaryBand,  
round(100 * Avg(Case When OverTime = 'No' Then Attrition = 'Yes' End),1) as No_Overtime_pct,  
round(100 * Avg(Case When OverTime = 'Yes' Then Attrition = 'Yes' End),1) as Yes_Overtime_pct  
From fact_attrition  
Group by SalaryBand  
Order by field(SalaryBand, 'Low', 'Medium', 'High', 'Very High');
```

	SalaryBand	No_Overtime_pct	Yes_Overtime_pct
►	Low	20.5	38.5
	Medium	16.9	42.4
	High	19.4	31.0
	Very High	6.2	13.9

Query 7: Attrition by Department & Job Role

- Compares employee attrition across different departments and job roles.
- Identifies the departments and job roles with highest attrition rates.

	Department	JobRole	Headcount	Attrition_rate_pct
►	Research & Development	Research Scientist	134	24.6
	Human Resources	Human Resources	54	24.1
	Research & Development	Healthcare Representative	107	22.4
	Research & Development	Laboratory Technician	121	22.3
	Sales	Sales Representative	84	21.4
	Sales	Sales Executive	104	18.3
	Research & Development	Manufacturing Director	116	18.1
	Sales	Manager	106	17.9
	Research & Development	Research Director	131	17.6
	Human Resources	Manager	43	14.0

```
Select
j.Department,
j.JobRole,
count(*) as Headcount,
round(100 * Avg(Attrition = 'Yes'),1) as Attrition_rate_pct
From fact_attrition f
Join dim_job j on j.JobID = f.JobID
Group by j.Department, j.JobRole
Order by Attrition_rate_pct Desc;
```

Query 8: Highest-Risk Job Role within Each Department

- Identifies the job role with the highest attrition risk within each department.
- Helps pinpoint specific job roles that require greater retention attention.

	Department	JobRole	headcount	Attrition_rate_pct	risk_rank_in_dept
►	Human Resources	Human Resources	54	24.1	1
	Human Resources	Manager	43	14.0	1
	Research & Development	Research Director	131	17.6	1
	Research & Development	Laboratory Technician	121	22.3	1
	Research & Development	Research Scientist	134	24.6	1
	Research & Development	Healthcare Representative	107	22.4	1
	Research & Development	Manufacturing Director	116	18.1	1
	Sales	Manager	106	17.9	1
	Sales	Sales Representative	84	21.4	1
	Sales	Sales Executive	104	18.3	1

```
With Role_rate as (
Select
j.Department,
j.JobRole,
count(*) as Headcount,
round(100 * Avg(Attrition = 'Yes'),1) as Attrition_rate_pct
From fact_attrition f
Join dim_job j on j.JobID = f.JobID
Group by j.Department,j.JobRole
)
Select
Department, JobRole, headcount, Attrition_rate_pct,
Rank() Over(PARTITION BY Department) as risk_rank_in_dept
from Role_rate
Order by Department, risk_rank_in_dept;
```

Query 9: High-Risk Employee Segment

- Identifies the employee segment with the highest attrition risk.
- Helps the business understand which type of employees are most likely to leave.

Result Grid	Filter Rows:	Export:	Wr
	High_risk_Headcount	pct_of_workforce	Attrition_rate_pct
▶	116	11.6	44.0

```
With High_risk as (  
  Select  
    EmployeeID,  
    Attrition  
  from fact_attrition  
  Where OverTime = 'Yes'  
  AND SalaryBand In ('Low','Medium')  
  AND TenureBand = '0-2 yrs'  
)  
  
select  
  (Select count(*) from High_risk) as High_risk_Headcount,  
  round(100*(Select count(*) from High_risk) / (Select count(*) From fact_attrition),1) as pct_of_workforce,  
  round(100* Avg(Attrition = 'Yes'),1) as Attrition_rate_pct  
from High_risk;
```

Query 10: Job Satisfaction & Work-Life Balance vs Attrition

- Compares attrition rates across Job Satisfaction and Work-Life balance levels.
- Identifies whether lower satisfaction or poor work-life balance is associated with higher attrition.

Result Grid				Filter Rows:	
	WorkLifeBalance	Headcount	Attrition_rate_pct		
▶	1	279	21.9		
	2	229	21.0		
	3	255	19.2		
	4	237	19.0		

```
Select  
  JobSatisfaction,  
  count(*) as Headcount,  
  round(100* Avg(Attrition = 'Yes'),1) as Attrition_rate_pct  
from fact_attrition  
Group by JobSatisfaction  
Order by JobSatisfaction;  
  
Select  
  WorkLifeBalance,  
  count(*) as Headcount,  
  round(100* Avg(Attrition = 'Yes'),1) as Attrition_rate_pct  
from fact_attrition  
Group by WorkLifeBalance  
Order by WorkLifeBalance;
```

Query 11: Tenure, Income & Age Comparison by Attrition

- Compares tenure, income and age between employees who stayed and those who left.
- Identifies differences in employee profiles associated with attrition.

	Attrition	Avg_Tenure	Avg_Income	Avg_Age	Avg_commute_distance
▶	No	4.2760	5889.4266	37.7792	7.0263
	Yes	2.3990	4327.0345	33.7488	10.2266

```
Select
f.Attrition,
Avg(f.YearsAtCompany) as Avg_Tenure,
Avg(f.MonthlyIncome) as Avg_Income,
Avg(e.Age) as Avg_Age,
Avg(e.DistanceFromHome) as Avg_commute_distance
from fact_attrition f
Join dim_employee e ON e.EmployeeID = f.EmployeeID
Group by f.Attrition;
```

Power BI Execution Dashboard Integration

 **Interactive Filtering :** Built dynamic slicers in Power BI allowing stakeholders to drill down by department and salary brackets.

 **KPI Cards & Trends :** Highlighted key workforce metrics including headcount, total attrition, attrition rate, high-risk employees, and average monthly income to give HR leadership quick view of workforce health.

 **Visual Storytelling :** Transformed raw SQL output rows into clear bar charts and distribution curves to communicate retention risks instantly to HR leadership.

